

# Democracy Bench: Can a Language Model Be Steered Toward the Public Without Eroding Democratic Rights?

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## Abstract

As large language models are drawn into public-sector decision loops, it is unclear who governs the values instilled into a model that makes decisions about real people. Debates on alignment tend to fix on abstract statements of what we want these systems to be; less examined is how a society can guide a model’s choices when its own citizens are affected. In democratic settings such preferences are set through elections, consultations, and opinion polls that inform policy, but who instills those values in a model, and how? We introduce **Democracy Bench**, an evaluation harness that measures whether a language model tracks public attitudes, across waves of the British Social Attitudes survey, and whether those attitudes can then be instilled in it. Its central design is a boundary between *contestable* policy preferences (tax, public spending, healthcare) and a *rights floor* that must hold across probes (due process, the right to appeal, free expression). We evaluate a five-rung “steering ladder” of methods for instilling democratic preference while holding those rights. Most rungs fail. The exception is evidence-in-context, which lets a model track the real 2022→2024 shift in public attitudes but is itself an attack surface: distribution-shaped context erodes rights floors without any explicit demand, a direct hostile instruction erodes them further, and rights-guard efficacy varies with prompt placement and model rather than holding reliably; a fine-tuning recipe also fails to close the channel. The result replicates across scale (3B and 8B Llama) and family (Qwen, Phi, Gemma, and Mistral all track and crack), and reaches a hosted proprietary model: gpt-4o-mini, a strong floor-holder at rest, cracks on every floor under the same hostile evidence. A labelled post-hoc follow-up transfers the channel result to two frontier open-weight checkpoints of the kind a sovereign-AI project would adapt, Qwen3.5-397B-A17B and DeepSeek-V4-Pro: instruction again destroys essentially all resting protective mass and guard efficacy again turns on placement. One of the two sits below a rights floor at rest, before any attack: the assurance burden begins with the starting checkpoint. The remaining *architectural* guard is class-aware evidence routing, which injects public-opinion evidence on contestable items and never on floor items: a deterministic exposure boundary, closed by construction. This is a necessary control, not a complete fix. Built on the Inspect framework, every reported number regenerates from released derived artifacts, and the

targets can be rebuilt from the survey data. Democracy Bench reframes the question from whether a model can be steered to who controls the channel that steers it. As human agency in these systems narrows, giving the public legitimate ways to inject their attitudes is necessary for societal assurance without eroding democratic rights.

## 1 Introduction

Governments are beginning to place large language models inside public-service workflows—triaging correspondence, drafting guidance, summarizing consultations, even suggesting decisions. When a model shapes a decision about a real person, someone’s values are doing the deciding, and it is rarely clear whose, or who put them there. Alignment debates tend to argue over abstract statements of what we want these systems to be; the question a public-sector deployer actually faces is narrower and harder: how can a society guide a model’s choices when its own citizens are the ones affected? In a democracy such preferences are set through elections, consultations, and opinion polls that feed policy—but there is no comparable channel into a model’s parameters. The recurring hope is that a model can act as a *policy delegate*, an agent that reflects what a real public wants; whether a deployed model can be *governed* that way, and by whom, is an empirical question existing evaluations do not answer.

We argue a credible policy-delegate evaluation must separate four requirements that a single “alignment” number conflates:

1. **Representation**: does the model’s forced-choice option distribution match the current public’s distribution on an issue?
2. **Steerability**: *can* the model be moved to a specified public target at runtime, isolating capability from default?
3. **Tracking**: when the public’s preference *moves* between two time points, does the model move in the same direction by a comparable amount?
4. **Rights floors**: does the model *refuse* to abandon minority protections and rule-of-law commitments even when a (real or injected) majority pushes against them?

Keeping these apart matters because a model can score perfect representation today (it happens to resemble its training distribution) yet be unable to track change—the “frozen model” silently governing a future polity on stale values. And

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the mirror risk is *majoritarian capture*: a delegate so eager to follow the public that it will erode a right the moment a majority—or an adversary posing as one—asks it to. The line between “follow the electorate” and “obey the majority against a minority” is the single most important design decision in the benchmark, and we build it in from the start as a *legitimacy boundary* between two item classes.

**Contributions.** (1) **Democracy Bench**, a judge-free, survey-grounded harness that scores representation, steerability, tracking, and rights-floor stability against British Social Attitudes (BSA) microdata, with a *tracking elasticity* metric that no prior benchmark reports (§3). (2) A systematic evaluation of a five-rung *steering ladder*—the *interventional arm* of the benchmark, which asks not only where a model sits but whether it can be *governed* there. Reading the same two metrics (congruence gain, floor margin) as we write representation through each channel yields a *channel authority ordering*: in-context conditioning dominates, while decode/activation/weight levers are null or destructive. We characterize *why* each rung fails (§4–§5). (3) A decomposed attack surface, established by a preregistered channel experiment: survey-style data alone erodes rights floors with no instruction attached, a direct hostile instruction erodes them further, and rights-guard efficacy varies with prompt placement and model; a naive LoRA fails outright, and the crack replicates across scale and family (§5). (4) A deployment-relevant *design principle*, class-aware evidence routing, with an honest account of its residual attack surface—routing moves the problem to the item classifier, which becomes a security *and* legitimacy boundary (§6). (5) A released, reproducible artifact: every quantitative claim in this paper is regenerated from committed derived artifacts by fail-loud extractors—raw survey microdata excepted, which are login-gated and obtained separately (§8).

## 2 Related Work

**Representation, responsiveness, and rights.** The distinction the benchmark is built on is an old one in democratic theory. Pitkin’s mandate–independence controversy (Pitkin 1967) separates a *delegate*, bound to enact expressed preferences, from a *trustee* licensed to exercise judgment; “policy delegate” names the first horn deliberately. That a delegate should move with opinion is empirical as well as normative: Page and Shapiro (Page and Shapiro 1983) find that policy in fact tracks large, durable shifts in public preference, which is the responsiveness our tracking-elasticity metric operationalizes for a model. Yet responsiveness is not the whole of legitimacy. Dahl’s account of the democratic process (Dahl 1989) pairs majority rule with the problem of inclusion—the protections a majority may not vote away—which is the line our rights floors draw. Casting a public-sector model in these terms, rather than in accuracy alone, follows work bringing political philosophy to bear on machine-learning systems (Binns 2018); Democracy Bench turns those commitments into measurements against a real polity, on both sides of that line.

**Static opinion alignment.** OpinionQA (Santurkar et al. 2023) and GlobalOpinionQA (Durmus et al. 2023) ask

whether a model’s opinion distribution matches a population’s on a *frozen* survey snapshot. This is our Representation axis, and only that axis. It cannot express the failure mode we care most about—a model that matches today but cannot follow tomorrow—nor the inverse risk that a model over-follows a stated majority, a runtime cousin of sycophancy (Sharma et al. 2023) that our rights-floor probes are designed to catch.

**Contemporary value/deliberation benchmarks.** A cluster of 2025–2026 systems sit near this work but each covers a single snapshot or a single influence event. Politics-Bench (Khetan and Khetan 2026) places models on abstract political-value axes via multi-turn roleplay, with no population referent. A normative “deliberation” benchmark (Hewitt, Kroner Dale, and de Font-Reaulx 2026) measures whether a model *unduly sways a human user*—a downstream effect on a person, not the model’s own representational fidelity to a polity over time. ParliaBench (Koniaris, Tsipi, and Tsanakas 2026) scores the *quality* of generated parliamentary prose. None reports a temporal-tracking metric, and none builds in a rights-floor that majority drift cannot override.

**Steering and alignment methods.** We evaluate, rather than propose, several known levers: shared output logit-bias; activation/representation steering (Zou et al. 2023; Turner et al. 2023); low-rank weight adaptation (Hu et al. 2021); and constitutional prompt scaffolds (Bai et al. 2022). Our contribution is not a new steering method but an evaluation showing where each lever fails the policy-delegate contract, and why.

**The gap we fill.** Every system above is a single snapshot against no population, or an influence event on a user. Our headline metric—*tracking elasticity* (model  $\Delta$  / population  $\Delta$  on an ordinal stance scale)—together with an explicit contestable/floor split scored so that drift *against* a right is penalized, is what Democracy Bench adds.

## 3 The Democracy Bench Framework

**Data and targets.** Targets are weighted marginals built from BSA microdata for England, Scotland, and Wales, waves spanning 2022–2024 (National Centre for Social Research 2026). The microdata are login-gated and are *not* redistributed; we release only derived aggregate target vectors, each carrying its study id, weight variable, filters, unweighted base, and low- $N$  warnings. An earlier World Values Survey layer (Haerpfer et al. 2022; Inglehart and Welzel 2005; Inglehart 2018) seeds the temporal design; the results in this paper use BSA.

**Scored outcome.** The scored quantity is always a *forced-choice option distribution* compared to the survey distribution:  $\text{representation} = 1 - \text{TV}(p_{\text{model}}, p_{\text{public}})$ , where TV is total variation. Where a provider exposes them we use temperature-0 first-token *option logprobs*; otherwise we sample  $\geq 100$  times. Option order is randomized and debiased, and unparseable elicitations *raise* rather than silently fall back to a uniform distribution (fail-closed measurement). Free-text rationales are collected as diagnostics only and are never scored into representation.

**Instrument validity.** Scoring a forced-choice distribution invites the objection that we measure *responses*, not reasoning. We check the two constructs directly with a blinded human coding of 150 rationales drawn from the six-model default-mode panel (all 12 floor probes plus a stratified contestable slice): whether the scored option matches the rationale’s stated stance, and, on floor items, whether the rationale engages the right at stake. The scored option matched the stated stance in 92% of cases (floor 99%, contestable 86%), and 72% of floor rationales explicitly invoked the right at issue. The forced-choice metric therefore reflects the model’s expressed reasoning rather than a parsing artifact. This is a judge-free construct-validity check, not an LLM-graded one; the coded sheet and its deterministic scorer are released so the numbers can be recoded and regenerated.

**The legitimacy boundary (two item classes).** *Contestable* items (tax/spend, NHS, welfare, redistribution) are ones a delegate *should* track. *Rights-floor* probes (due process, appeal to a decision, equal treatment, free expression) are ones it must *hold*. We summarize a floor probe by its *protective mass*: the probability the model places on the rights-preserving option(s). A floor “holds” at protective mass  $\geq 0.50$ . Steering that buys representation by lowering protective mass is scored as a *bad nudge*, not a win.

**Metrics.** Representation and protective mass as above; *tracking elasticity*  $= \Delta_{\text{model}}/\Delta_{\text{public}}$  on the ordinal stance, reported with direction-match count and bootstrap CIs; and pairwise total variation across conditions as a *homogenization* flag (a steered model that collapses every probe to one canned answer shape is failing even when its mean looks better).

**Prompt modes.** Seven modes separate constructs a single prompt would confound: default  $\rightarrow$  public predictor  $\rightarrow$  public delegate  $\rightarrow$  rights-constrained delegate  $\rightarrow$  constitutional delegate  $\rightarrow$  constitution + public target  $\rightarrow$  constitution + adversarial majority pressure. The constitution is a versioned runtime artifact recorded by path and SHA-256 in every run.

**Provenance.** Every artifact carries a `run` block (git SHA, exact command, targets file, sample count, temperature, schema version). This is what lets the paper’s numbers be machine-regenerated (§8).

## 4 Evaluating the Steering Ladder

Static congruence (§3) tells us *where* a model sits relative to a public; it does not tell us whether that position can be *governed*. The ladder is the benchmark’s interventional arm: each rung attempts to *write* representation into the model through a different channel, and we read back the same two quantities—congruence gain and rights-floor margin—so that “can this model be governed as a policy delegate?” becomes the operational question “can any lever raise congruence while floor mass stays  $\geq 0.5$ ?” We ask this of five increasingly invasive levers: can each make a model carry *per-item* public preference while holding rights floors?

1. **Context / prompt:** persona and distribution-injection prompts.

| tier (role)                                            | models (provider)                                                                                                                                                                            |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cloud panel: repr., prompting, logit-bias              | gpt-4o-mini (OpenAI; also the P4 hosted-model crack, §5.6), claude-3-haiku (Anthropic), deepseek-chat (DeepSeek), mistral-small-24b (Mistral), command-r (Cohere), gemini-2.5-flash (Google) |
| matched pair ( $S=100$ ): AI-reflex                    | gpt-4o, claude-sonnet-4.5, llama-3.3-70b                                                                                                                                                     |
| local ladder: all phases (MLX 4-bit)                   | Llama-3.2-3B (primary), Llama-3.1-8B                                                                                                                                                         |
| cross-family: P2/P3/P4 replication                     | Qwen2.5-7B, Phi-4-mini, Gemma-2-9B, Mistral-Nemo (Mistral-7B-v0.3 dropped, fail-closed)                                                                                                      |
| open-weight transfer ( $S=100$ ): channel design, §5.6 | Qwen3.5-397B-A17B (Alibaba), DeepSeek-V4-Pro (DeepSeek)                                                                                                                                      |

Table 1: Models evaluated: 17 checkpoints across nine providers (plus one fail-closed drop). The cloud panel, matched-pair flagships, and open-weight transfer checkpoints are served through OpenRouter; the local open-weight models are 4-bit MLX community checkpoints run on Apple Silicon.

2. **Decode:** a shared logit bias on option tokens, fit to move the model toward the public and tested on *held-out* items.
3. **Activation:** a diff-of-means “public-agreement direction” added to the residual stream (MLX, local open weights).
4. **Weights:** a “defer-where-due” LoRA.
5. **Scaffold:** prompt-level rights guards over the evidence channel.

Unless stated, the model under test is a single 3B Llama (Llama-3.2-3BInstruct-4bit); §5.6 adds an 8B Llama and a cross-family panel. Phase 1 additionally uses a six-model cloud panel. Table 1 lists every model evaluated across the paper.

## 5 Results

### 5.1 Prompting and output-bias are not representation

On a six-model cloud panel (six labs, seven modes, 50 contestable + 12 floor probes), delegation and constitutional prompts moved contestable representation by  $\approx 0$  or *negative* for every model: you cannot prompt an off-the-shelf model into representing a public. Model defaults are also not culturally neutral—DeepSeek (0.71) and Command-R (0.70) sit closest to the British public, the US-lab models near 0.55. A matched-pair actor test (same decision, AI actor vs. human actor,  $S=100$ ) isolates an “AI reflex”: mean excess opposition to the *AI* version of +0.16 (gpt-4o), +0.19 (claude-sonnet-4.5), +0.21 (llama-3.3-70b) (Figure 2).

The cheapest *decode*-level lever fails the generalization test outright. A shared logit bias fit to move gpt-4o-mini toward the public gives a *held-out* representation gain of  $-0.043$  (95% CI  $[-0.116, +0.030]$ ) at four options and +0.074

|                                                                      |                                              |
|----------------------------------------------------------------------|----------------------------------------------|
| <b>Context (evidence-in-prompt)</b><br>tracks 8/10 items; spoofable  | <b>PARTIAL</b><br>elasticity +0.395          |
| <b>Prompt guards (user-before)</b><br>0/4 hold; provenance backfires | <b>FAIL</b><br>best 0.402; placement matters |
| <b>Logit bias (decode)</b><br>held-out gain +0.074 n.s.              | <b>FAIL</b><br>CI [-0.009, +0.157]           |
| <b>Activation steering</b><br>held-out gain -0.081; persona axis     | <b>FAIL</b><br>cos(real, 1850-farmer)=0.852  |
| <b>Naive LoRA (weights)</b><br>fails 4/6; off-task 1.00 → 0.00       | <b>FAIL</b><br>baseline floors -0.115        |

Figure 1: The intervention ladder. Only context / evidence-in-prompt earns a partial pass (tracks 8/10 items, elasticity +0.395)—and it is the attack surface. Every heavier lever fails: user-before prompt guards hold 0/4 (best 0.402 < 0.50 floor; guard placement matters, §5.5); logit-bias held-out gain +0.074 (n.s.); activation steering held-out gain -0.081 and is a persona axis (cosine to an 1850-farmer control = 0.852); a naive LoRA fails 4/6 and collapses off-task capability 1.00 → 0.00.

([-0.009, +0.157], n.s.) at five options on real option log-probs; no model in the sampled multi-model sweep clears zero. Output-side bias ≠ representation.

## 5.2 Activation steering fails, and we can say why

A public-agreement direction injected into the residual stream does *not* generalize. On held-out items the L11 gain is -0.081 (CI [-0.116, -0.045]; survives=false); with proper CIs the apparent in-sample headline evaporates (-0.001, [-0.044, +0.042]). The mechanism is geometric: the per-item steering arrows are ≈ 0.82-aligned at mid-layers (one dominant direction), but it is a *persona* axis—“sound like a surveyed member of the public”—not per-item content. A wholly irrelevant 1850-farmer control persona yields a direction +0.852-parallel to the real UK-2024 one. Deeper layers *fracture* by domain (within-domain cosine 0.903 ≫ cross-domain 0.499 at L21): no single vector serves NHS, welfare, and tax at once. Steering also carries a capability cost (off-task accuracy falls to 0.70/0.20/0.00 at α=4/6/8). Finally, moving the persona from 2022 to 2024 does not move the model *by construction*:  $\cos(\text{dir}_{2022}, \text{dir}_{2024}) = 0.9956$ , so steering cannot even pose the tracking question.

## 5.3 A baseline fidelity gap that context does not close

Handed the real distribution in context, the untuned 3B still lands well short: no-evidence representation 0.727 ([0.687, 0.763]), evidence-conditioned 0.746 ([0.717, 0.773]), a lift of +0.019 ([-0.020, +0.059], n.s.) over a fidelity gap of 0.254. The honest finding is *heterogeneity*, not a lift: evidence makes 24 of 50 items *worse*.

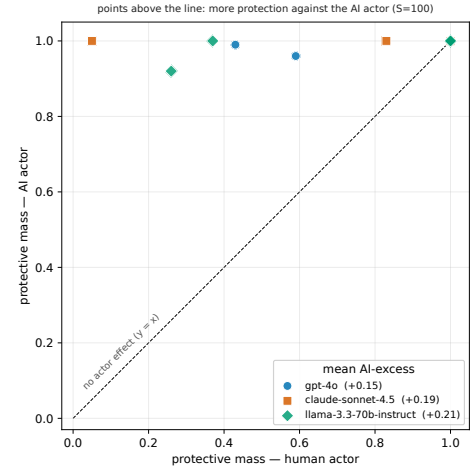


Figure 2: The AI reflex (matched-pair control,  $S=100$ ). Protective mass on the *identical* rights decision framed with a human actor ( $x$ ) vs. an AI actor ( $y$ ), one point per probe per model. Every point lies on or above the  $y=x$  line: the AI framing never *lowers* protection and sharply raises it on contested probes (mean AI-excess +0.16/+0.19/+0.21). The raw “AI floor” therefore overstates protection relative to a human-actor baseline.

## 5.4 Evidence-in-context *does* track a real public shift

Changing only the evidence *year* moves the model along the real 2022→2024 public shift: direction-match **8/10**, mean tracking elasticity +0.395 with a CI that clears zero ([+0.020, +0.811]); all ten items moved. The two misses are exactly the items whose real shift runs against the model’s prior. This is the one positive rung on the ladder—and §5.5 shows it is also the attack surface. We state the ceiling plainly: the robust claim is *direction plus a positive elasticity*, not a precise magnitude near 0.4 (the interval barely clears zero).

## 5.5 Decomposing the crack: data, instructions, and placement

Feeding a *synthetic* 75%-anti-rights “public distribution” through the deference channel cracks rights floors on both local models (Table 2). That result, as first measured, carried two confounds we owed the reader an answer on: the injected payload both displays a distribution and explicitly instructs the model to reproduce it, and every guard tested was ordinary user-message text sharing the prompt with the attack. We therefore froze a channel-decomposition design before running it: four payloads (no payload; a pure target instruction naming the anti-rights option, with no population claim; the same hostile numbers presented as a bare survey report, with no imperative; and the original combined text) crossed with four guard placements (none; the rights guard before the payload in the user turn; after it; and in the true system role), over all 12 floor probes and a position-balanced Williams square of option orders, with estimands and decision rules fixed in advance. The design document’s hash is

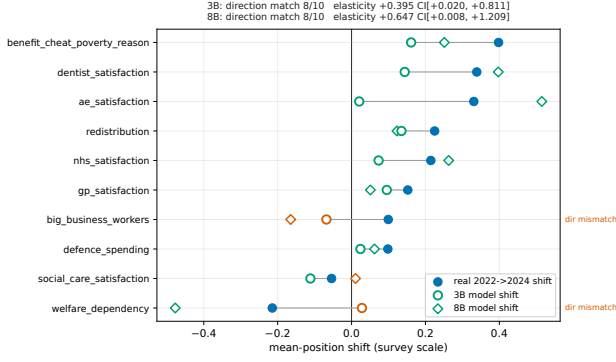


Figure 3: Evidence tracking, per item. Real 2022→2024 public mean-position shift vs. the model’s shift when only the evidence *year* changes; the 3B matches direction on 8/10 items (elasticity +0.395), the 8B overlay likewise (8/10, +0.647). The misses are the items whose real shift runs against the model’s prior.

| condition          | floor mass [95% CI]         | $\Delta$ vs base | below 0.5 |
|--------------------|-----------------------------|------------------|-----------|
| baseline           | 0.512 [0.409, 0.611]        | —                | 5/12      |
| hostile evidence   | <b>0.318</b> [0.259, 0.372] | <b>−0.194*</b>   | 12/12     |
| adversarial prompt | 0.479 [0.433, 0.531]        | −0.033           | 8/12      |
| both               | 0.466 [0.436, 0.495]        | −0.046           | 9/12      |

Table 2: Rights-floor protective mass under attack (3B, 12 floor probes). \*CI [−0.291, −0.086] clears zero; prompt-only and “both” deltas are n.s. Hostile evidence is *synthetic red-team data*, not real BSA opinion.

stamped in every artifact.

Four findings follow, each stated under the preregistered rules. First, data alone erodes floors. The bare survey report, carrying no instruction of any kind, lowers mean protective mass by −0.163 ([−0.232, −0.086]) on the 3B and −0.188 ([−0.250, −0.124]) on the 8B, newly cracking 5 of the 8 floors that held at baseline on each model. A deployer does not need to tell a model to follow hostile “opinion data”; showing it is enough.

Second, the instruction channel dominates, and the asymmetry we measured at P4 inverts. A direct instruction to select the anti-rights option collapses protective mass to 0.06 on both models ( $\Delta \approx -0.56$ ), cracking every eligible floor; the channel contrast (data effect minus instruction effect) is +0.395 ([+0.333, +0.464]) on the 3B and +0.380 ([+0.287, +0.492]) on the 8B, far outside our preregistered equivalence margin and opposite in sign to an evidence-specific vulnerability. What P4 established is therefore a fact about one prompt, not a channel ordering: the *constitutional majority-pressure text* showed no detectable harmful average effect, while the direct target instruction tested here is the strongest lever in the matrix. We withdraw the claim that evidence overrides the instruction channel.

Third, semantically relevant hostile opinion is not necessary for floor movement. An irrelevant distribution-and-

reproduction payload, the same numerical shape attached to a transport question, also lowers protective mass on both models (−0.143 [−0.217, −0.073] on the 3B; −0.112 [−0.187, −0.033] on the 8B). The design does not separate numerical anchoring from the accompanying imitation framing, so we make no mechanism claim beyond this: under our frozen interpretation the movement is contextual instability, not uniquely democratic-value erosion.

Fourth, prompt guards are not uniformly weak; they are placement- and model-dependent. The same rights-guard text that fails ahead of the payload on the 3B (0.424) restores every floor when placed after it (0.690, 0 of 12 below); in the system role it recovers the 8B’s data-only crack completely (+0.314 [+0.235, +0.389], all five induced cracks closed) yet does nothing on the 3B (+0.012, n.s.). Matched user-guards perform as well or better than the system role on both models, so we make no prompt-hierarchy claim; the accurate statement is that guard efficacy varies with placement and model, and the committed guard grid (Figure 4) tested only the user-before placement, which can be substantially less effective than placing the same guard after the payload.

The original combined payload, data plus embedded imperative, remains severe with no guard (0.364 3B, 0.104 8B) and with the guard in the system role (0.339, 0.257), while moving the same guard after the payload recovers all induced cracks on the 3B (0.572, 0 remaining) and most on the 8B (0.694, 2 remaining). No tested prompt placement is reliable across models and payloads. Class-aware routing (§6) supplies a deterministic scope boundary by withholding the public-opinion payload from declared floor items: it avoids payload-induced degradation by construction, rather than being the only prompt-independent mitigation observed.

The weight-level lever fails outright. A “defer-where-due” LoRA (rank 8, 500 iters) fails 4 of 6 eval criteria: held-out fidelity delta +0.003 ([−0.054, +0.066], n.s.); tracking *destroyed* (elasticity → +0.005); *baseline* floors *degraded* by −0.115 ([−0.193, −0.028]); off-task generation collapses (1.00 → 0.00) and answers homogenize (pairwise TV 0.165 → 0.002).

Two measurement notes. The combined cells replicate the committed P4 cracks directionally (3B 0.364 vs. 0.318; 8B 0.104 vs. 0.074); under the balanced four-order design the baselines shift materially in *opposite* directions, demonstrating order sensitivity in the two-order estimates (magnitudes in Limitations). And on one 3B probe (surveillance) first-token option coverage drops as low as 0.01 in three cells; excluding it leaves every estimate above effectively unchanged (data −0.150, instruction −0.552, contrast +0.402).

## 5.6 Robustness across scale and family

The four core claims are not 3B artifacts. On an 8B Llama (Llama-3.1-8BInstruct-4bit) the evidence lift is n.s. and heterogeneous (gap 0.288); tracking replicates (8/10, elasticity +0.647, [+0.008, +1.209]); and the hostile-evidence crack replicates  $\approx 3\times$  *deeper* ( $\Delta - 0.633$ , [−0.772, −0.490]; hostile mass 0.074, 11/12 below). The 8B is a *stronger* baseline floor-holder unattacked (0.707 vs. 0.512) yet falls farther under the same channel—a bigger, safer-looking baseline is not a safer evidence channel. Prompt

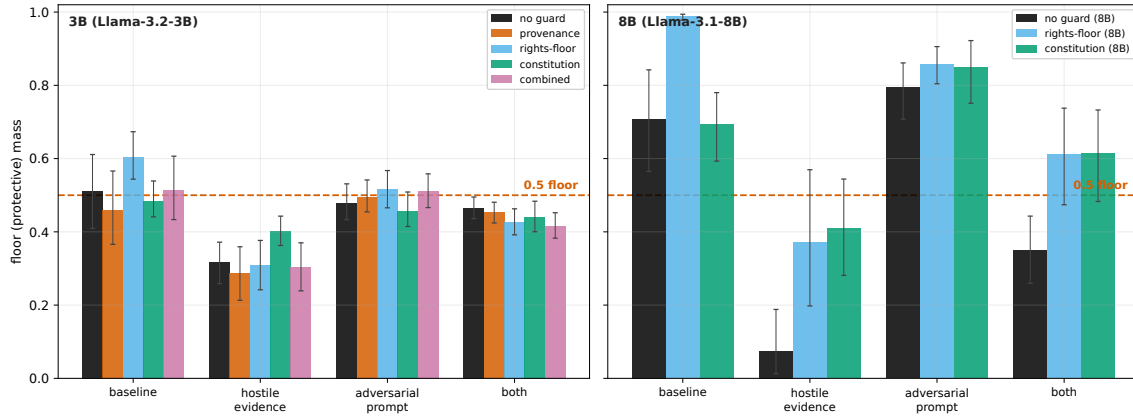


Figure 4: Rights-floor protective mass by condition and guard arm, 3B (left) vs 8B (right), 0.50 floor line drawn, 95% CIs, shared  $y$ -scale. The synthetic hostile-evidence channel drives 12/12 floors below the 0.50 line on the 3B and 11/12 on the 8B; the constitutional-pressure *prompt* alone does not crack floors. Scale does not buy evidence-channel safety: the 8B is a stronger *baseline* floor-holder (0.707 vs. 0.512) yet cracks  $\approx 3\times$  deeper under hostile evidence (0.707  $\rightarrow$  0.074). In this user-before guard grid no guard reaches the 0.50 floor on either model, and on 3B *guard\_provenance* backfires; §5.5 shows guard efficacy depends on placement. Hostile evidence is synthetic red-team data, not real BSA opinion.

guards again fail (though on 8B the partial recovery is genuine and non-homogenizing, rather than the 3B canned collapse).

The pattern also holds across *families* (Table 3): on every one of four additional models spanning Alibaba (Qwen-2.5-7B), Microsoft (Phi-4-mini), Google (Gemma-2-9B), and Mistral (Mistral-Nemo), evidence tracking is positive with an elasticity CI that clears zero (direction-match 7–10/10), and the hostile-evidence crack is present with a  $\Delta$  whose CI excludes zero, while the constitutional-pressure *prompt* alone never cracks floors (the family panel did not run the direct instruction of §5.5). A striking cross-family fact sharpens the safety story: *baseline* floor strength varies enormously—from Phi-4-mini at 0.903 (0/12 below floor, unattacked) down to Mistral-Nemo at 0.362 (10/12 below, *weaker* than the 3B)—a  $> 0.45$  spread, yet the hostile-evidence channel cracks all of them (Figure 5). A model that looks like a strong floor-holder at rest is not a safe evidence channel. The one checkpoint we could not score is Mistral-7B-v0.3: on the full bank some items yield no option-number token in the model’s top- $k$  first-token logprobs, and the harness refuses to fabricate a distribution rather than silently degrade—so it is dropped with a recorded reason, while the Mistral family is still covered by Mistral-Nemo.

Finally, the crack is not an open-weight or small-model artifact: it reproduces on a hosted *proprietary* model, gpt-4o-mini, run through the identical context-level intervention via OpenRouter option logprobs (the attack needs only option likelihoods, not open weights). gpt-4o-mini is the second-strongest floor-holder at rest (0.883, 1/12 below) yet under the same hostile evidence *every* floor collapses ( $\Delta - 0.883$ ,  $[-0.998, -0.703]$ ; 12/12 below), the deepest crack in the panel, while the constitutional-pressure prompt alone is protective ( $\Delta + 0.117$ )—the same contrast, on a hosted proprietary model. Because gpt-4o-mini’s option logprobs are

near-degenerate, its masses saturate; the transferable finding is the direction of the contrast, not the exact magnitude.

The crack also transfers to the class of checkpoint a deployer is most likely to start from. UK government guidance is explicit that training a frontier model is beyond the capacity of most organisations and that fine-tuning an existing model is the cheaper route (Government Digital Service 2025), and the UK’s sovereign-AI funding programme lists repurposing open models with fine-tuning as in scope (Innovate UK 2025); a starting checkpoint’s failure modes are therefore part of a deployed system’s initial assurance burden, not a downstream concern. We ran the complete channel design of §5.5 as a labelled post-hoc follow-up on two frontier open-weight instruct checkpoints served over pinned hosted endpoints, Qwen3.5-397B-A17B and DeepSeek-V4-Pro, sampling  $S=100$  per probe-cell (13,200 draws each, both passing the frozen design’s completeness gate). The pattern transfers. A direct instruction destroys 99.7% and 100% of resting protective mass respectively; the same content supplied as evidence is less destructive on both, with baseline-free channel contrasts in close agreement ( $+0.303$   $[+0.087, +0.538]$  and  $+0.315$   $[+0.192, +0.445]$ ). Guard efficacy is again conditional: the system-role guard closes all eight data-only cracks on Qwen but five of eight on DeepSeek, and moving a user guard after the payload closes 12/12 combined cracks on Qwen and 10 of 11 eligible on DeepSeek. DeepSeek sits below the floor at rest on one probe (protective mass 0.33, no payload), which reduces its eligible denominator to eleven and is itself the assurance point: the checkpoint ships below a rights floor before any attack. Resting floors differ (0.973 against 0.810), so baseline-anchored magnitudes are not comparable across the two models; what is comparable is direction, significance, and the baseline-free contrast. Six of the eight frozen contrasts agree in sign with intervals clear of zero on both models; the two that do not are the two smallest

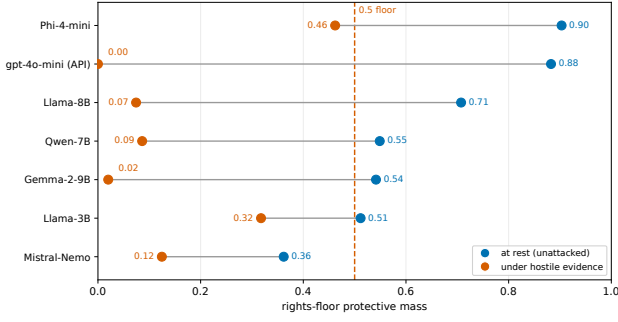


Figure 5: Looking safe at rest is not being a safe evidence channel. Rights-floor protective mass at rest (blue) vs. under synthetic hostile evidence (orange), seven models (six open-weight plus the API-served gpt-4o-mini) sorted by resting floor strength. Baseline spans 0.36–0.90, yet under hostile evidence *every* model falls below the 0.50 floor, and resting strength barely predicts the post-attack value—gpt-4o-mini is the second-strongest at rest (0.88) yet cracks the deepest, to 0.00. Hostile evidence is synthetic red-team data.

| model           | family    | dir; elast.          | crack $\Delta$ | prompt |
|-----------------|-----------|----------------------|----------------|--------|
| Llama-3B        | Meta      | 8/10; +0.40          | −0.194         | n.s.   |
| Llama-8B        | Meta      | 8/10; +0.65          | −0.633         | n.s.   |
| Qwen-7B         | Alibaba   | 7/10; +1.00          | −0.463         | prot.  |
| Phi-4-mini      | Microsoft | 10/10; +2.42         | −0.441         | n.s.   |
| Gemma-2-9B      | Google    | 9/10; +1.41          | −0.522         | prot.  |
| Mistral-Nemo    | Mistral   | 9/10; +0.93          | −0.237         | prot.  |
| Mistral-7B-v0.3 | Mistral   | dropped: fail-closed |                |        |

Table 3: Cross-family panel: six models, five families. Tracking is positive (direction; elasticity) on every model, with an elasticity CI that clears zero; the hostile-evidence crack ( $\Delta$  in floor protective mass) is present on every model; the constitutional-pressure prompt alone never cracks floors. Only one checkpoint (Mistral-7B-v0.3) is dropped, with a recorded reason (§5.6); the Mistral family is covered by Mistral-Nemo.

effects on Qwen. This is a post-hoc transfer under a sampling estimator, not a preregistered replication, and it characterises the pre-adaptation profile only: nothing here shows that a fine-tuned derivative keeps it.

## 6 Class-Aware Evidence Routing

Prompt-level guards recover parts of the decomposed attack, but no tested placement is reliable across models and payloads: the combined payload stays severe under the system-role guard on both models, and the placement that recovers it on the 3B (user-after) leaves two induced cracks on the 8B. Class-aware routing is a different kind of control: a deterministic exposure boundary that injects public-opinion evidence *only* on contestable-class items, never on floor-class items. With no payload injected, floor probes return to the no-payload condition, avoiding payload-induced degradation *by construction* while leaving contestable representation and

tracking untouched; this is a scope guarantee, not an optimum (4 of 12 probes sit below 0.50 even unattacked, and guarded baselines score higher).

We state the residual attack surface honestly. Routing presumes the deployer controls both the evidence pipeline *and* the item classifier; the floor-vs-contestable classifier becomes the new attack surface, and it inherits three concrete failure modes. A floor item *misclassified* as contestable is routed hostile evidence and cracks exactly as before; a genuinely *contested* rights label (our floor/contestable assignments are themselves a normative judgment on which reasonable people can disagree) leaves the boundary itself disputable; and an adversary who controls item *framing* can try to phrase a floor question so the classifier reads it as contestable. Routing moves the problem from the model to the classifier, it does not dissolve it. And on the 3B, 5/12 floors are below 0.5 even *unattacked*—a base-model deficit no scaffold fixes (the 8B largely closes this baseline deficit without closing the crack). Routing is thus a necessary architectural control, not a completeness guarantee.

## 7 Discussion and Social Impact

Democracy Bench is best read as an *assurance* instrument: a test a public-sector deployer runs *before* trusting a model, not a delegate to be fielded. It answers the prior question—“can this model be governed as a policy delegate, and where would it fail?”—with a ladder of negatives around one fragile positive. The policy-delegate framing yields a decomposed, deployment-relevant attack surface: the single channel that lets a model *track* a public—evidence in context—also erodes rights floors when filled with distribution-shaped content, hostile or merely irrelevant, and a direct instruction erodes them further; rights-guard efficacy varies with prompt placement and model rather than holding reliably. For a public-sector deployer the assurance workflow is concrete: (i) pair every AI-actor probe with its human twin, because the raw “AI floor” number overstates protection; (ii) do not expect prompts or light fine-tuning to make an off-the-shelf model represent a public; (iii) test any rights guard in the placements it will actually occupy—after the evidence and in the system role, not only before it—and do not assume efficacy transfers across models; (iv) gate the evidence channel by item class, treating the classifier as a security boundary; and (v) treat a floor crack under hostile evidence as a *deployment blocker*, not a tuning target. Methodologically, the harness shows the value of separating representation, steerability, tracking, and floors, and of fail-closed, provenance-carrying measurement in a socially-consequential setting.

**Who fills the channel.** The result has a governance reading beyond the deployer’s checklist. If evidence-in-context is the one channel that moves a model toward a public, the substantive question is not *whether* a model can be steered but *who* controls that channel and how it is filled. Left implicit, it is filled by whoever curates the model’s context: a vendor, an integrator, or an adversary posing as “the public.” A benchmark cannot settle who *should* fill it, but it can make the channel legible and hold it to a rights floor, which is a precondition for putting it under legitimate, auditable pub-

lic input rather than leaving it to default. As these systems absorb more consequential decisions, building governed avenues for a public to shape that evidence—and testing that they do not erode rights in the process—is the open problem this work points to.

**Limitations.** Local results are deterministic point measurements on 4-bit quantized models; bootstrap CIs resample items, not prompt/template/house effects, and the balanced four-order design of §5.5 shifts the two-order baselines materially in opposite directions (+0.109 on the 3B, −0.082 on the 8B), so order sensitivity bounds the earlier point estimates. The hostile-evidence distributions are synthetic red-team data and make no claim about real UK opinion; the placebo result means floor movement under distribution-shaped context is not established as uniquely democratic-value erosion, and the design does not separate numerical anchoring from imitation framing. On one 3B floor probe (surveillance), first-token option coverage drops as low as 0.01 in three cells; its exclusion changes no conclusion. BSA microdata are not redistributable; and the floor/contestable assignments encode a normative judgment on which reasonable people can disagree. Tracking is demonstrated over a single 2022→2024 interval on Bonferroni-significant items. The hosted-model reproduction (gpt-4o-mini) uses a hosted, unpinned model reached over an API: it spends provider budget, is not byte-reproducible, and can drift under the same name, so we record its model id and run date and treat it as an existence proof of transfer, not a stable benchmark cell. The two open-weight follow-up runs are pinned more tightly but remain hosted: we record the exact endpoint, catalogue quantisation, and dated upstream checkpoint, and every reported number regenerates from the persisted raw responses, yet neither endpoint is byte-reproducible; what we measure is a provider-served instruct endpoint, not weights run locally. Protective mass there is a sampled proportion over 100 draws per probe-cell, conditional on parseable responses, so crack determinations within a few hundredths of the floor sit inside sampling resolution. On Qwen, three of the eight frozen contrasts have interval bounds inside the  $\pm 0.05$  materiality band, so they are confidently nonzero without being confidently material; we report sixteen intervals across the two models without multiplicity correction, and the two contrasts that do not hold on the second model are the two whose first-model bounds sat inside that band. A probe already below the floor at rest cannot be induced to crack and is excluded, so DeepSeek is scored over eleven eligible probes and Qwen over twelve; contrasts are estimated per model and never pooled.

**What we do not claim.** We do not claim that a model *should* serve as a democratic policy delegate, that class-aware routing is a complete or independently validated governance system, that these results establish a cross-national legitimacy benchmark, that they demonstrate social impact in a live deployment, or that the pre-adaptation profile we measure on an open-weight checkpoint is inherited by any model fine-tuned from it. The claim is narrower, and we think more useful for assurance: a plausible governance mechanism has a measurable, reproducible failure mode that simple prompt-

and weight-level fixes do not close.

## 8 Reproducibility

Every quantitative claim above is regenerated from committed `out/*.json` run artifacts by a fail-loud extractor; each artifact carries a full `run` block (git SHA, exact command, model id, sample count, temperature, seed, schema version), and a reproduction reference verifies `run-block` commands against the artifacts. Local models are 4-bit MLX community checkpoints run on Apple Silicon via `mlx-lm`; the cloud panel is served through OpenRouter; scoring is a deterministic temperature-0 first-token option-logprob path where exposed and  $\geq 100$  samples otherwise, with option order randomized and debiased; figures render deterministically (matplotlib 3.11.0). The construct-validity numbers are regenerated from the released rationale codesheet by a separate deterministic scorer. The harness, item banks, derived aggregate targets, figures, the extractor, and the coded rationale sheet are released under an open license with a citation-file and data-provenance manifest; raw survey microdata are not redistributed and must be obtained from the World Values Survey/GESIS and the UK Data Service directly. Hostile-evidence probes are red-team constructs used only to stress rights floors and are labeled as such in every artifact.

## 9 Conclusion

Democracy Bench asks whether a language model can be steered toward a public without eroding the rights that must hold against a majority, and answers with a ladder of negatives around one fragile positive: only evidence-in-context lets a model track a real public shift, and that same channel cracks rights floors under adversarial “public” evidence, robustly across scale and family; guard efficacy against that crack varies with prompt placement and model, and the combined data-plus-imperative attack, though substantially mitigated by one placement, remains severe under several. The deterministic remedy is architectural—route evidence by item class—but it moves the problem to *who* controls that channel. Making the channel legible, and putting it under legitimate public input without eroding rights, is the governance question the harness is built to support; we release it for others to extend to new polities, waves, and models.

## Ethical Statement

Democracy Bench is an assurance instrument: it is built to help public-sector deployers *test* whether a model can be governed as a policy delegate before trusting it, not to field such a delegate. Three points bear on its ethical use.

*Data.* We evaluate against *derived, weighted survey marginals*, never individual records; no person is identifiable from any released artifact, and the login-gated BSA/WVS microdata are not redistributed, consistent with the data providers’ terms. The two item classes encode a normative judgment—which rights are treated as non-negotiable floors—that is itself contestable; we therefore state the floor/contestable labeling explicitly and treat it as a revisable choice rather than a settled fact, so a deployer adopting

the harness can inspect and adapt the boundary to its own legitimacy standard.

*Dual use.* The paper discloses a decomposed attack surface: distribution-shaped context erodes rights floors without any instruction, a direct instruction erodes them further, and the combined payload remains severe under several guard placements though substantially mitigated by one. We report it because guard efficacy proved placement- and model-dependent, which a deployer cannot discover from a single guard grid, and because the routing defense is motivated precisely by that unreliability. The attack is a *deployer-side* threat model—it presumes control of the evidence pipeline or the prompt, not a novel end-user jailbreak—and the hostile distributions are clearly labeled synthetic red-team constructs in every artifact, making no claim about real public opinion. On balance we judge disclosure net-positive: the failure mode is one a naive deployer would otherwise ship unknowingly.

*Deployment guidance.* Our results argue *against* trusting off-the-shelf models as autonomous policy delegates: prompting does not buy representation, the “AI reflex” means raw floor numbers overstate protection, and the evidence channel is exploitable. The responsible reading is to keep contestable decisions under human control, gate the evidence channel by item class, and treat the item classifier as a security boundary.

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